

YourSchool (TeachHub) ek advanced, full-stack multi-tenant School Management Platform hai. Is system ko schools, academies, teachers, students aur super-admins ke pure academic aur financial workflows ko automate karne ke liye design kiya gaya hai. Is document me Technical Stack, Payment Architecture, Authentications, Login Workflow aur Sabhi Roles (Super Admin, Admin, Teacher, Student, Pending User) ke features ki poori jankari di gayi hai.

Portal Type	Live URL Link	Description (■■■■■)
Main Website	yourschoolacademy.com	Main Landing Page, School Landing pages, Public Features & General Information.
Student Login	student/login	Dedicated login portal for Students to view attendance, report cards & fees.
Teacher Login	teacher/login	Portal for Teachers to mark attendance, upload study notes & enter marks.
Admin Login	admin/login	School Admin/Principal dashboard for fee setup, user approvals & payroll.

TeachHub modern web and mobile tech stack par build kiya gaya hai jo high scalability, security, aur real-time data synchronization provide karta hai:

Category	Technologies Used	Role & Functionality in App
Backend Stack	Node.js, Express.js (v5) MongoDB & Mongoose ODM	RESTful API Server, high performance asynchronous request handling, flexible MongoDB schema database for multi-school tenant data.
Authentication & Security	JWT (JSON Web Tokens), Bcryptjs, Sanitize-HTML	Secure token-based auth with HTTP authorization headers, password hashing using bcrypt, and strict XSS HTML sanitization for text inputs.
Real-time & Media	Socket.io (v4), Cloudinary API, Multer	Real-time WebSockets for instant user approval notifications & status changes. Cloud storage integration for images, avatars, study notes & documents.
Push Notifications & Email	Firebase Admin SDK (FCM), Nodemailer	Cross-platform Firebase Push Notifications sent to Android native apps. Automated email alerts for password reset & OTPs.
Frontend Web Stack	React 19, Vite Engine, Tailwind CSS 3, Lucide Icons	Ultra-fast Single Page Application (SPA), responsive UI styled with Tailwind CSS, modular component architecture.
Routing & Data Viz	React Router DOM 7, Recharts, Axios	Client-side dynamic routing with protected role layout guards. Interactive visual analytics charts for attendance, revenue & grades.

TeachHub me ek robust **Hybrid Dual Payment Architecture** use kiya gaya hai jo online direct transactions aur offline manual verification dono support karta hai:

Agar student cash deposit karta hai ya direct QR code scan karke transfer karta hai, to app me offline-request submit kiya jata hai (with UTR Number, payment screenshot, notes). Admin Dashboard me alert aata hai, jaha Admin receipt check karke approve-offline ya reject-offline kar sakta hai.

- **Student School Fees:** Admin dynamic *FeePlan* (Tuition Fee, Admission Fee, Transport, Exam Fee) create karte hain. Cycles: Monthly, Quarterly, One-time. Students fee status track karke pay karte hain.
- **School SaaS Subscription:** School Admin Superadmin ko TeachHub platform access service subscription pay karta hai. Superadmin free-trial period bhi grant kar sakta hai.
- **Teacher Payroll & Salaries:** Admin sabhi teachers ke monthly compensation structure set karte hain, total payout amount mark-paid karte hain aur pending salary ledger balance maintain rehta hai.
- **Payment Audit Logs & Receipts:** Har ek financial transaction ka complete audit trail (PaymentAuditLog model) maintain hota hai. Instant downloadable branded PDF receipts generate hoti hain (e.g., TH-2026-XXXXX).

Step	Workflow Phase	Technical Mechanism & User Experience
Step 1	User Credentials Entry	User specific page (/student/login, /teacher/login, /admin/login) par email/password enter karta hai ya Google Auth use karta hai.
Step 2	JWT Token & Role Verify	Backend credentials verify karke JWT Access Token returning karta hai. User ka role (student, teacher, admin, superadmin) aur requestStatus fetch hota hai.

Step 3	Status-Based Routing Guard	• Agar requestStatus === 'pending' -> Directed to Pending Approval Page. • Agar requestStatus === 'approved' & role === 'student' -> Student Dashboard. • Agar requestStatus === 'approved' & role === 'teacher' -> Teacher Dashboard. • Agar role === 'admin' -> Admin Dashboard. • Agar role === 'superadmin' -> Super Admin Control Panel.
Step 4	Real-Time WebSockets Sync	Socket.io active connection establish rehta hai. Jab Admin pending user ko approve karta hai, tab user ko page reload kiye bina automatically Live Dashboard me redirect kar diya jata hai.

5. Detailed Feature Breakdown by User Role (■■■ ■■■■■ ■■ ■■■■■■)

A. Pending Approval Page Features (■■■■■■■ ■■■ ■■ ■■■■■■)

Jab tak School Principal ya Admin new Student/Teacher ki registration request approve nahi kar deta, user **Pending Page** par rehta hai. Features include:

- **Live Registration Status Banner:** Real-time status display karta hai (e.g. 'Pending Approval', 'Entrance Test Scheduled', 'Under Review').
- **Requested School & Role Display:** Show karta hai ki user ne kis school aur kis role (Student ya Teacher) ke liye apply kiya hai.
- **Online Admission Exam Portal:** Agar school ne entrance exam mandatory kiya hai, to student issi page se online entrance exam de sakta hai. Exam complete hone par status 'exam_completed' update ho jata hai.
- **Socket.io Real-time Listener:** Admin ke approve/reject action karte hi bina refresh kiye instant dashboard redirect hota hai.
- **Request Re-submission & School Change:** User galat school choose karne par request modify ya school change kar sakta hai.

B. Student Portal Features (■■■■■■■ ■■■ ■■ ■■■■■■)

Approved Students ke liye dedicated academic & fee portal:

- **Personal Student Dashboard:** Class, Section, Roll Number, Daily Attendance Percentage, Pending Fees summary, aur Upcoming Exams alert.
- **Online Fee Payment & History:** Tuition, Exam aur Transport fee breakdown dekhna, Razorpay online gateway se pay karna ya Cash UTR upload karna, aur branded PDF payment receipts download karna.
- **Academic Results & Report Cards:** Term-wise (Unit Test, Half Yearly, Annual) marks, grades, percentage, class merit rank, aur complete PDF Report Card download.
- **Digital Study Materials & Notes:** Teachers dwara upload kiye gaye Subject Notes, PDF assignments, syllabus tracking documents view/download karna.
- **Attendance History Calendar:** Month-by-month interactive calendar view (Present, Absent, Leave indicators).
- **Mobile Native App (APK):** StudentApp native Android app push notifications ke saath for instant exam & homework alerts.

C. Teacher Portal Features (■■■■ ■■■ ■■ ■■■■■■)

Teachers ke daily classroom management & grading tools:

- **Class & Subject Schedule:** Assigned classes, sections aur daily subject timetable ka structured display.
- **Digital Attendance Marking:** Single-click daily student attendance take karna (Present / Absent / Late mark karna).
- **Exam Marks & Grading Entry:** Exam terms ke mutabiq students ke marks input karna with auto-grade calculation.
- **Study Notes & Media Upload:** Subject-wise notes, PDF documents, Cloudinary uploaded files aur YouTube lecture links publish karna.
- **Syllabus Progress Tracker:** Subject syllabus modules mark completed karna aur visual completion percentage metrics dekhna.
- **Salary & Compensation Ledger:** Monthly salary status, payout history aur salary receipts view karna.

D. Admin Portal Features (■■■■■ / ■■■■■ ■■■■■■■■■■ ■■ ■■■■■■)

School Directors aur Principals ke liye full institutional management control panel:

- **School Branding & Public Profile:** School name, logo, principal photo, contact details, address, description aur banner management.
- **User Approval & Admission Desk:** Pending Student aur Teacher applications verify karna, approve/reject karna, ya entrance exam schedule karna.
- **Fee Plan Engine & Structure:** Custom Fee Plans (Monthly, Quarterly, One-Time) create karna, class-wise fee apply karna.
- **Offline Payment Verification:** Cash / Manual UPI offline payment proofs review karna, UTR check karke approve/reject karna.
- **Teacher Payroll Management:** Teachers ke base salary & compensation set karna, payout mark karna, salary ledger maintain karna.
- **Academic Year & Class Management:** Academic session set karna, Classes (e.g. Class 1 to 12), Sections (A,B,C), aur Subjects assign karna.
- **Exams, Results & Merit Roll Generator:** Exam schedule create karna, marks verify karna, automatic Merit Roll rank generate karna aur Report Cards issue karna.
- **Notice Board & Push Broadcast:** School-wide circulars, notices aur mobile push notifications send karna for events & holidays.

E. Super Admin Portal Features (■■■■■ ■■■■■■ ■■ ■■■■■■)

Global Platform Owner ke liye top-level multi-school management system:

- **Multi-Tenant School Management:** Platform par new schools register karna, school details edit karna, activate/deactivate ya delete karna.
- **Platform SaaS Subscription Control:** Schools ke liye SaaS subscription pricing set karna, Razorpay/Offline payment collection, free period/trial grant karna.
- **Global Payment Audit Log:** System me hone wale har ek payment transaction ka centralized financial audit log (who paid, amount, school, timestamp, status).
- **User Role Delegation & Password Reset:** System-wide kisi bhi user ka role elevate/change karna, password reset, account deletion management.
- **Global System Broadcast:** All schools, admins, teachers aur students ko universal announcement or alert push notification bhejna.

- **Platform Analytics & Server Health:** Total schools, total students enrolled, active revenue, database diagnostics & performance metrics.

6. Summary Comparison Matrix (■■■■■■■ ■■ ■■■■■■■■■■■■ ■■■■■■)

Feature Module	Pending User	Student	Teacher	School Admin	Super Admin
Approval Status / Gate	✓ View	-	-	✓ Approve All	✓ Master Control
Attendance	-	✓ View Own	✓ Mark Daily	✓ Full Report	✓ Global Analytics
Fee / Subscription	-	✓ Pay Online/Offline	✓ Salary View	✓ Setup & Verify	✓ SaaS Revenue
Report Cards & Marks	✓ Exam Test	✓ Download PDF	✓ Enter Marks	✓ Generate Merit	-
Study Materials	-	✓ Download Notes	✓ Upload PDF/Media	✓ Monitor	-
Multi-School Management	-	-	-	✓ School Profile	✓ Full Platform

Document compiled automatically for TeachHub / YourSchool Platform. All URLs & features verified from live codebase and production deployment at <https://yourschoolacademy.com/>.